torch.eye#

https://pytorch.org/docs/stable/generated/torch.eye.html

Description:

Returns a 2-D tensor with ones on the diagonal and zeros elsewhere. n (int) – the number of rows m (int, optional) – the number of columns with default being n out (Tensor, optional) – the output tensor. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()).

Function Signature

```
torch.eye(n, m=None, *, out=None, dtype=None,
layout=torch.strided, device=None, requires_grad=False) →
Tensor#
```

```
Parameters
```

```
Keyword Arguments
```

```
Returns
```

```
Return type
```

Parameters

Keyword

out (Tensor, optional) – the output tensor. dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

Returns:

Tensor

Code Examples

```
>>> torch.eye(3)
tensor([[ 1.,  0.,  0.],
        [ 0.,  1.,  0.],
        [ 0.,  0.,  1.]])
```

Detailed Documentation

Documentation

Returns a 2-D tensor with ones on the diagonal and zeros elsewhere.

n (int) – the number of rows

m (int, optional) – the number of columns with default being n

out (Tensor, optional) – the output tensor.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()).

layout (torch.layout, optional) – the desired layout of returned Tensor. Default: torch.strided.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

A 2-D tensor with ones on the diagonal and zeros elsewhere